

UNSTRUCTURED DATA FOUNDATIONS

Study Guide — What's Inside a Document, and How Foundry Sees It

Week 4 · Day 1 · Prepared by Rajesh Pasham

Program Context

Item	Detail
Session Type	Concepts and architecture lecture — no live Palantir platform work today
Platform	Individual Palantir Developer Tier accounts, used from Day 3 onward
This Week's Lab (Day 3)	Insurance Document Intelligence Pipeline — extracting and linking real claim, policy, and medical documents to your existing Ontology

Where Today Fits

Every object you've built since Week 1 has a property like `fileReference` or a linked Claim Document — but until now, nothing in your workflows has actually looked at what's inside those files. Week 4 is about that gap: turning a PDF, a scanned form, or a photo into something your Ontology and your AI workflows can actually use.

Learning Objectives

By the end of today, you will be able to:

- Explain what a media set is, and why it's a first-class Foundry concept
- Know the difference between raw text extraction and OCR, and why picking the wrong one silently produces bad results
- Understand what metadata a document carries beyond its text
- See how an extracted document connects back to the Ontology objects you already built

1. What Counts as Unstructured Data

Anything that doesn't fit neatly into rows and columns:

- Documents — claim forms, policy PDFs, medical records, police reports
- Images — photos of vehicle or property damage, scanned handwritten forms
- Audio and video — recorded claim calls, inspection walkthroughs

2. Media Sets: A First-Class Concept

A media set is a collection of media files that share a common schema — for example, all PDFs, or all vehicle-damage photos. It is not just a folder: media sets carry the same governance used since Week 1 (Markings, security policies, versioned uploads) and support schema-specific operations directly, like OCR or format-based filtering. Foundry recognizes six schema types: document, image, audio, video, spreadsheet, and DICOM (medical imaging) — each with its own available operations. Today's lab works primarily with document and image.

3. Media References: Using Data Without Copying It

- A media reference points to a file, rather than duplicating it — the same principle as a Link Type from Week 1: you reference the relationship, not a second copy of the data
- Media references can live as a column in an ordinary dataset — associating a PDF with its file name, page count, and extracted text as additional columns, tabular data sitting alongside unstructured content
- Media reference object properties bring this into the Ontology — this is what lets a Claim Document object actually display and preview its underlying file in Object Explorer or Workshop

4. Document Classification

Media sets can be filtered by type and format — for example, narrowing a mixed media set down to just PDFs before running a PDF-specific operation. Classification is what makes batch processing possible: a pipeline that handles claim forms, medical records, and photos together needs to know which is which before applying the right extraction method to each.

5. Text Extraction: Raw Text vs. OCR

The single most important decision in this topic — and an easy one to get wrong silently.

Method	When to Use It
Raw Text Extraction	For born-digital PDFs — documents where the text is already embedded and selectable. Fast, and the extracted text is exactly what's in the file.
OCR (Optical Character Recognition)	For scanned documents and images with no embedded text layer — a photo of a form, a faxed document. OCR reads the visual characters and reconstructs text, which can introduce errors.

Why the distinction matters: running raw text extraction on a scanned document returns nothing useful. Running OCR on a born-digital PDF works, but slower and with unnecessary risk of misread characters. Picking correctly matters for both speed and accuracy.

6. Auto Mode: Letting the Platform Decide

Auto mode inspects each page individually, determining page by page whether text can be extracted directly or whether OCR is required — useful when a single PDF mixes born-digital pages with scanned inserts. Auto mode isn't free of judgment calls; it's a default worth checking, not a guarantee. For a batch of claim documents where quality is inconsistent, Auto is usually the safer starting point. For a known, uniform document type, an explicit choice can be faster.

7. Metadata Extraction

Every media item carries metadata beyond its content: file name, format, page count, upload date — available directly, without running any extraction at all. Metadata is often enough to route a document before extraction even runs: a claim document's page count or format alone can flag it as incomplete or the wrong type, before spending compute extracting its text.

8. Content-Based Search

Once text is extracted, it can be searched like any other text field — “find every claim document mentioning water damage” becomes a real query, not a manual review task. This is the foundation Day 2 builds on: content-based search here is exact-match, while Day 2 introduces semantic and hybrid retrieval, which finds conceptually similar content even without matching words.

9. Connecting Documents Back to the Ontology

Your Claim Document object type has had a fileReference property since Week 1 — today's concepts are what finally make that property mean something. Wednesday's lab links extracted text and metadata from real claim forms and medical records back to the Claim and Insurance Policy objects you've been working with since Week 1, using the exact media reference object properties introduced in Section 3.

10. Common Pitfalls

- Assuming every PDF needs OCR — a born-digital PDF already has embedded text; running OCR on it wastes compute and risks introducing errors that raw extraction wouldn't have
- Treating a media set like an ordinary dataset — media sets have their own governance and operations; the same Markings and security discipline from Week 1 still apply, but the mechanics are specific to media
- Skipping metadata because extraction seems more important — metadata is often the cheapest, fastest way to classify or route a document, before spending compute on full text extraction
- Forgetting that a media reference is a reference, not a copy — the same discipline from Week 1's Link Types; duplicating file content defeats the purpose and multiplies storage cost

11. Glossary

Term	Meaning
Media Set	A collection of media files sharing a common schema, with its own governance and schema-specific operations
Media Reference	A pointer to a media item, usable in a dataset column or an Ontology object property, without copying the underlying file
Raw Text Extraction	Extracting embedded, already-selectable text from a born-digital document
OCR (Optical Character Recognition)	Reading visual characters from a scanned document or image and reconstructing text
Auto Mode	A text-extraction setting that decides per page whether raw extraction or OCR is required
DICOM	A schema type for medical imaging media sets

12. Further Practice

Continue into Day 2, where these foundations extend into chunking strategies, embedding models, and hybrid retrieval — the bridge into Day 3's hands-on Insurance Document Intelligence Pipeline.